RAJALAKSHMI ENGINEERING COLLEGE

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(An Autonomous Institution, Affiliated to Anna University, Chennai)
Department of Artificial Intelligence and Data Science
V Semester B. Tech AIDS

AD23532 – Principles of Data Science

Academic Year: 2026-2027 **Batch: 2024-2028**

Ex3a: Model Planning and Building - Classification

Learning Objective:
The objective of this experiment is to enable students to plan, build, and evaluate basic machine learning models using real-world datasets.

Problem Statement:
A telecommunications company has noticed that a growing number of customers are discontinuing its services, resulting in reduced revenue and increased customer acquisition costs. After completing data preprocessing and exploratory analysis, the company now aims to develop predictive models that can identify customers who are likely to churn. Early prediction of customer churn will enable the company to implement targeted retention strategies, improve customer satisfaction, and reduce business losses.

As a Data Scientist, your task is to build and compare multiple classification models to predict whether a customer will churn. You will train different machine learning algorithms, evaluate their performance using appropriate classification metrics, and identify the most effective model for customer churn prediction. Finally, interpret the important factors influencing customer churn and communicate your findings through suitable visualizations and performance comparisons.

Tasks to Perform:

1. Use CustomerChurn_Preprocessed.csv file which you created in Ex2.
2. Split the cleaned dataset into Training (80%) and Testing (20%) datasets.

Model Building

3. Train a Logistic Regression classifier and predict the customer churn on the test dataset.
4. Train a Decision Tree Classifier and predict the customer churn on the test dataset.
5. Train a Random Forest Classifier and predict the customer churn on the test dataset.

Model Evaluation

6. Evaluate each model using Confusion Matrix, Accuracy, Precision, Recall, F1-Score.
7. Compare the performance of all three models using a single bar chart showing the evaluation metrics.
8. Identify the best-performing model based on the evaluation results.

Business Recommendation

9. Based on the best-performing model, provide three recommendations that the telecommunications company can implement to reduce customer churn.